

Supervised Learning Foundations

Instructor Solutions

Book 2 Lesson 1 marking guide and model reasoning

Mbaye Kebe | SRAI | Controlled review candidate | 10 September 2026

This guide provides expected reasoning, not only final answers. Award full credit when the learner states the information boundary, preserves the evaluation design, and qualifies conclusions appropriately.

Marking overview

Section	Points	Evidence for full credit
A	20	Correct definitions and distinctions
B	20	Correct calculations and interpretations
C	30	Valid leakage and evaluation reasoning
D	30	Reproducible professional evidence

1–5 Core concepts

An observation is the defined unit; features are admissible inputs; the target is the future quantity or class; prediction time fixes admissibility. Training performance is optimistic because fitting adapts to training data. Validation supports choices; the untouched test supports final estimation. Leakage may be target, temporal, group or pipeline leakage. A baseline must face the same target, timing and test cases.

6 Empirical risk

The expression averages a declared loss over n training pairs. y_i is the observed target, $f(x_i)$ the prediction, and L the consequence assigned to the error. It estimates performance on the training sample and does not equal expected risk unless the transfer assumptions are justified.

7 MAE and RMSE

MAE averages absolute errors in target units. RMSE squares errors before averaging and therefore emphasizes large misses. RMSE is especially relevant when extreme staffing shortfalls or high-demand misses are disproportionately costly.

8 Improvement

Absolute improvement = $40.496 - 30.876 = 9.620$ RMSE units. Percentage improvement = $9.620 / 40.496 \times 100 = 23.76\%$, approximately 23.8%.

9 ROC AUC

ROC AUC summarizes ranking across thresholds. The operating threshold depends on prevalence, intervention capacity and the relative costs of false positives and false negatives. Calibration and threshold-specific precision and recall must also be examined.

10–14 Case analysis

The post-outcome field is target or temporal leakage. Reject RMSE 2.048 and retain 30.876 as the controlled estimate under the chronological split. Use chronological train, validation and final test periods. Compare training-mean and seasonal-naive baselines; examine MAE, RMSE, residuals and high-demand error; examine region and demand-level subgroups. Communicate incremental value with uncertainty and explicit deployment limitations.

15–19 Professional evidence

A complete submission records environment, seed 42, split identifiers, rerun status and changed design choice. Threshold analysis must connect precision, recall and intervention volume. Deployment evidence includes data quality, calibration, subgroup behavior, latency and override monitoring. Warning and suspension triggers name owners, evidence windows and recovery requirements.

Scoring rubric

Criterion	Excellent	Partial	Insufficient
Task definition	Unit, target, horizon and action explicit	One element unclear	Task reduced to algorithm
Evaluation	Split reproduces intended use	Split plausible but weakly justified	Training or random score treated as proof
Leakage	Failure and control correctly identified	Leakage named without operational test	Leaked result accepted
Interpretation	Claims qualified by uncertainty and scope	Some qualifications missing	Deployment inferred from score
Reproducibility	Complete rerun and evidence record	Partial record	No executable evidence